METEOROLOGY: This can be defined as the study of atmosphere and its phenomena. It is a science that deals with laws and principles as they apply to atmospheric phenomena. It is a science of atmosphere and its attendant activities.

Meteorology can also be defined as a branch of physics that deals with physical process in the atmosphere that produce weather. Agricultural meteorology on the other hand is the branch of applied meteorology which deals with the response of crops to the physical environment. It is the study of physical process of the atmosphere that produce weather in relation to agricultural production.

CLIMATOLOGY: It is the science dealing with the factors which determine and control the distribution of climate over the earth's surface.

ATMOSPHERE: It is a colourless, tasteless and odourless mixtures of gases that surround the earth. It extends up to a height of about 1-600 Km. However 99% of total mass of the atmosphere is within 40 Km from the earth.

Composition of atmosphere include
i. Nitrogen, ii. Oxygen iii. Argon, iv. Carbon dioxide and other gases.

METEOROLOGICAL STATION: This is a place or a enclosure where all elements of weather are measured and recorded. Its dimension measured 10 metres by 6 metres in size.

THE MOST APPROPRIATE SITE FOR METEOROLOGICAL STATION:

The conditions necessary for the most appropriate site for meteorological station include:

1. It should be located on a levelled ground covered with short grasses.
2. It should not be sited on or close to a hills, in a depression or on steep slope.
3. It should not be sited also near tall trees or buildings.
4. Usually fenced around with ware gause for the security of the instruments and ensure free air circulation.

VARIOUS COMPONENTS OF ~~THE~~ METEOROLOGICAL STATION:

1. THE STEVENSON SCREEN: It is a compartments in the station where thermometers are kept. The thermometers kept in the station include maximum and minimum thermometer, wet and dry bulb thermometers. The screen is built to provide shade under which the shade temperature of the air can be measured.

2. <u>THERMOMETER</u>: Air temperature is measured manually with the aid of maximum and minimum thermometer and autographically by a thermograph.

The maximum thermometer is a mercury-in-glass thermometer which contain a small glass index. when the temperature rises the mercury expands and pushes the index along the tube. when the temperature falls, the mercury contracts leaving behind the index. The maximum temperature is read on the scale at the end of the index near the mercury.

The minimum thermometer is an alcohol-in-glass thermometre in which when the temperature rises the alcohol expends and flows past the index and when the temperature falls the alcohol contracts and pulls the index along the tube. The end of the index nearer the meniscus show the minimum temperature.

The instruments is reset by tilting or by using a magnet to draw back the index to the mercury.

The maximum and minimum ~~Ther~~ thermometer's are used at weather station to measure the highest and lowest temperature within the day respectively.

3. <u>THERMOGRAPH</u>: It is a self-recording thermometer used to measure air ~~the~~ temperature.

4. **HYGROMETER**: It is used in measuring relative humidity. The hygrometer comprises of wet and dry bulb thermometer placed side by side in the Stevenson screen.

5. **BAROMETERS**: This is an instruments used in measuring air pressure. The various types of barometers are mercury, aneroid barometers, the barograph and altimeter.

6. **Cup ANEMOMETER**: It is used in measuring wind speed. It consists of 3 or 4 cups, conical or hemispherical in shape mounted symmetrically about vertical axis. The metals cups which are fixed to the ends the arms rotate according to the prevailing wind speeds. The recording takes place in the cups.

7. **WIND VANE**: wind direction is observed with the aid of a wind vane which consists of a rotating arm pivoted on a vertical shaft. The arrow of the wind vane always points in the direction from which the wind blows and the wind is named after the direction.

8. **CAMPBELL - STOKES SUNSHINE RECORDER**: It used in measuring duration of sunshine.

9. EVAPORIMETER: This is an equipments generally used in measuring evaporation. The evaporimeter are of two basic types. The Piche evaporimeter and free water surface in tanks or pans.

10. EVAPOTRANSPIROMETER: used in measuring the evapotranspiration.

11. RAIN GAUGE: This is an instruments used in measuring rainfall. Autographic rain gauge include ~~intitling eg~~ i. tilting siphon ii. The tipping bucket iii. weighing collector system. Theses instruments depends on type or operating mechanism.

12. The instruments used in measuring ~~the~~ insolation are as follows:

i. NET RADIOMETERS: used in measuring only net radiation.

ii. PYRANOMETERS: used for measuring the total short wave radiation from the sky incident on a horizontal surface at the ground.

iii. PYRHELIOMETERS: used for measuring the solar intensity or the direct beam solar radiation at normal incidence. These are the most accurate of all radiation instruments.

iv. PYRRADIOMETERS: used for measuring only the albedo of the surface.

v. Albedometers - used for measuring only the albedo of the surface.

vi. PYRGEOMETERS: Used for measuring infrared long-wave radiation from the earth's surface or the atmosphere depending upon whether it is downward facing or upward facing.

13. SOIL THERMOMETER: Used in measuring soil temperature. The soil temperature is measured at various depths 5cm, 10cm, 20cm, 50cm and 100cm. The thermometer is mercurial with bulbs embedded in paraffin wax. The thermometer is suspended in steel tubes and inserted into the soil at various depths.

MAINTAINANCE OF METEOROLOGICAL STATION.

i. Building should not be located close to the meteorological station.
ii. Big trees should not be allowed to grow around the station.
iii. The grasses around should be kept low at all the time.
iv. Observation of the elements should be regular and prompt.
v. Animals should not be allowed access to the station particularly the pans and tanks in drier areas and small animals use the tanks for bathing.
vi. The station should be under key and lock to avoid theft and tampering.
vii. Malfunctioning equipments should be replaced promptly.

1st. Assignments

CLIMATE:

Climate is the accumulation of daily and seasonal weather events of a given location over a period of 30–35 years. The concept of climate is more than the average weather condition, it also includes weather events consideration of variabilities (departure from averages), extreme condition, and the probabilities of frequencies of occurrences of given weather condition.

Climate, on the other hand is refered to as the characteristics condition of the atmosphere deduced from repeated observation over a long period of time.

Weather is the state of the atmosphere at a given time and place.

Climatic condition are so natural that could not be changed by man. However, he can adopt his agricultural methods and technique to these condition and can change the microclimate by using irrigation, shelter belts, mulching, or provision of [illegible] when all animal [illegible]

[illegible]

appropriate [illegible]

CLIMATIC ELEMENTS

1. TEMPERATURE: This can be defined as the degree of hotness or coldness of a substance, determined by the extend of its molecular activity and is measured by thermometer. Air temperature, Solar radiation is the main source for air temperature, Solar radiation received at the outer limits of the earth's atmosphere is quite constant. However the amount that reaches the earth's surface is highly variable depending greatly on the amount of clouds in the atmosphere.

SOME TERMS IN RELATION TO TEMPERATURE:

DIURNAL AND SEASONAL VARIATION: It is a common knowledge that day temperature is higher than night temperature. The temperature variation between day and night is called Diurnal variation of temperature. Diurnal variation is less in coastal areas, while it is considerable in inlands. Temperature varies ~~with the~~ with season, higher temperature prevailing during summer and lower temperature during winter. The seasonal variation in temperature is less near the equator and is more at higher latitude.

TEMPERATURE GRADIENT: Temperature decreases as altitude increase. The decrease in temperature in higher altitudes in the air is known as vertical temperature gradient. The reason for the decline in temperature with decreased altitude are.

i. Air is heated by the earth.

2. Water vapour content of air decreases with increase in altitude reducing the ~~ca~~ capacity to hold heat.

3. Expansion of air at higher elevations.

LAPSE RATE: It is defined as the rate of change in temperature with altitude or the decrease in air temperature as one ascends into the atmosphere.

A regular air temperature decreases with an average rate of 0.65°C for every 100 metres of ascent. Vertical temperature decrease or gradient is expressed in lapse rate. Lapse rate varies very widely. The normal lapse rate is 6.5cm/Km

ATMOSPHERIC TEMPERATURE: These can be defined as the weight exerted upon the earth by the atmosphere. The average air pressure at sea-level is arround 1013 millibars but it varies slightly according to latitude. The air pressure decreases with altitude at 4500 metres.

ADIABATIC LAPSE RATE: The rate at which the temperature changes as air rises or fall. This rate is constant in dry air. The adiabatic lapse rate is 10°c/Km.

Units of Temperature:

Temperature is measured either in Fahrenheit or Celsius or in Kelvin Units.

In Fahrenheit Scale, the melting and boiling points of water are 32 and 210°F respectively. In Celsius scale they are 0 and 100°C. Absolute temperature is measured in Kelvin Units when there is no activity of molecules of a substance then its is said to be at 0°K. The melting point of water in Kelvin Scale is 273°K.

The Conversion formular.

$$C = (F - 32)\tfrac{5}{9}$$

$$F = C(\tfrac{9}{5}) + 32$$

$$K = C + 273.$$

2. THE ATMOSPHERIC PRESSURE:

The pressure exerted by the atmosphere on the earth's surface is called atmospheric pressure. Atmosphere contains air molecules which have a certain weight. Therefore, pressure is defined as the force per units area. The atmospheric pressure is defined as pressure exerted by a column of air with a cross-sectional area of a given units extending from the earth's surface to the uppermost boundary of the atmosphere. The atmospheric pressure is expressed in millibars.

Units of atmospheric pressure is millibars.

The average air pressure at sea-level is around 1013 millibars but it varies slightly according to latitude. The air pressure decreases with altitude; at 4502 m it is only 572 millibars. This is because at high altitude air is less dense.

3. HUMIDITY

Humidity is a measure of the amount of water in the atmosphere. Liquid water is converted into water vapour by evaporation for which necessary energy is provided by solar radiation in the form of temperature water vapour is colourless, odourless and tasteless and consists of minute droplets of water suspended in the air. The amount of water vapour in the atmosphere depends upon wind and temperature. Wind distribute the water vapour in the atmosphere. Higher the temperature more is the amount of water vapour that can be hold by the atmosphere. The air is said to be saturated when it holds maximum amounts of water vapour at a particular temperature. If the temperature rises the ~~tempe~~ atmosphere becomes unsaturated as it can accomodate some more water vapour.

The quantity of water vapour hold in the atmosphere at any particular time can be expressed as. 1. Absolute 2. Specific 3. Relative humidity.

Absolute humidity is the absolute or actual quantity of water vapour by weight presents in a given volume of air. The formular:

$$\text{Absolute humidity} = (g/m^3) = \frac{\text{weight of water}}{\text{volume of air}}$$

Specific humidity: It is the weight of water vapour per unit weight of air (including water vapour). Expressed in water vapour present in a 1 kilogram of air.

$$\text{Specific humidity } (g/kg) = \frac{\text{weight of water vap.}}{\text{weight of the air including water vapour}}$$

Relative humidity: This is the ratio between the amount of water vapour present in the air and the amount of water vapour required for saturation at a particular temperature and pressure. Expressed as percentage or ratio

$$\text{Relative humidity (RH)} = \frac{\text{water vapour present in the air}}{\text{water vapours required for saturation}} \times 100$$

Relative humidity is the most popularly used index for measuring air humidity. It is easily measured and indicates the degree of saturation of the air.

$$RH = \frac{\text{water vapour present in the air}}{\text{water vapour required for saturation}} \times 100$$

MASS MIXING RATIO or humidity mixing ratio: it is the mass of water vapour per kilogram of dry air.

4. RAINFALL: The word rainfall is often used as a Synonym with precipitation. The falling of any type of condensed moisture to the ground water is called precipitation. Rainfall is precipitation in the form of liquid drops larger than 0.5mm in diameter falling on the earth. When water droplets in a cloud grow in size and become heavy enough, they fall to the earth as rain. Ordinary rain drop size varies from 0.5 to 4 mm in diameter.

FORMS OF PRECIPITATION

• DRIZZLE - Uniform precipitation of very numerous minutes drops, it usually falls from fog or thick layer of Status.

RAIN - most common form of precipitation falls from rising air, when temperature at lower levels is above 0°c. The droplets, as a rule are larger in size than drizzle. The maximum size to which a rain drop can grow is about 5mm in diameter though occassionally rain drops may be as small in size as the drizzle drops.

Snow - precipitation of solid water mainly in the form of branch hexagonal crystals or stars - at temperature not far from freezing points. They are usually melted together in large snow flakes.

SLEET: precipitation consisting of melting snow or a mixture of snow and rain, snow and sleet may melt completely while falling through the air. They will then appear as rain on the ground.

HAIL - precipitation of ball or irregular lumps of ice of diameter ranging from 5.5mm or more - They are either transparent or composed of clear layers of ice alternating with opaque layers of snow-like structure. Hail falls almost exclusively in violent thunderstorms and in very rare at temperatures below freezing ~~at~~ the earths surface.

Rainfall is measured with rain gauge. Rainfall is generally stated in unit of inches, millimeters, and centimeters that fall per unit of time.

For meteorological reco[illegible] is reckoned by a [illegible]

with at least 0.25mm of rain recorded.
Autographic rain gauge include.
i. Tilting Siphon ii. The tipping bucket
iii. Weighing Collector System (W.M.O 1970).
This depends on type or operating
mechanism.

5. SOLAR RADIATION:

The Sun supply virtually all the energy (99.97%) received by the earth. Solar radiation is the source of the energy for all the physical processes taking place in the atmosphere. This energy also drives the process of photosythesis, evaporation, heating the soil and air. A Continues constant emition of solar radiation of about (1.94 Cal/cm^2/min) is refered to as solar Constant. It is defined as the energy falling in one minute on a surface area of one square Centimetre at the outer boundary of the atmosphere, held normal to the sunlight, at the mean distance of the earth from the Sun.

FORMS OF SOLAR RADIATION

1. DIRECT SOLAR RADIATION: It is the radiation received directly from the Sun by a surface normal to the incident radiation.
2. DIFFUSED or SKY RADIATION: It is the radiation Scattered by the suspended particles

The quantity of diffuse radiation depends on latitude season and cloudness.

3. <u>GLOBAL SOLAR RADIATION</u>: It is the sum total of direct and diffuse solar radiation received on a horizontal surface from the sun directly and from the sky as scattered radiation.

4. <u>REFLECTED SOLAR RADIATION</u> OR ALBEDO: It is a solar radiation reflected without any change in its quality.

5. <u>TERRESTRIAL RADIATION</u>: It is the thermal radiation emitted by the earth.

6. <u>NET RADIATION</u>: It is the radiation balance between global and reflected solar radiation.

Units: In International Standard (SI) units solar radiation is expressed as <u>watts per square metre</u>.

The metric (CGS) unit most commonly used in meteorology prior to adoption of SI units is calories per square centimeter per min. i.e one watt energy is equivalent to one joule per second.

1 watt = 1 joule/s

697.93 W/m^2 = 1 $Cal/cm^2/m$

10,000 Lux = 3.47 Meg joule

MICRO CLIMATE

The climate of a very small or restricted area especially when this differ from the climate of surrounding area. micro climate exists, for example near bodies of water which may cool the local atmosphere or in heavy urban areas where brick, concretes, asphalt ~~abs~~ absorb sun energy, heat up, and re-radiate that heat to the ambiant air the resulting in urban heat is kind of micro climate

IMPORTANCE OF MICRO CLIMATE

1 Micro climate can offer an opportunity as a small growing region for crop that cannot thrive in the broader area.

2 Micro climate can be used to the advantage of gardener who carefully choose and position their plants

THE IMPORTANCE OF MOISTURE IN AGRICULTURE

moisture play an important part in agriculture as it affects or determines plant growth and development. it availability, or scarcity, can mean a successful harvest or diminution in yield or total failure

major importance of moisture in plant — agriculture are as follows

1 water maintain cell turgidity for structure and growth
2 Transporting nutrient and organic compound through the plant.
3 moisture comprising much of the living protoplasm in cell
(4) moisture serves as a raw materials for various chemical process, such as photosynthesis and through transpiration buffering the plant against wide temperature fluctuation.

CLIMATE AND AGRICULTURAL DEVELOPMENT PLAN.

Climate affects agricultural production in diverse ways either directly or indirectly. Climate must therefore taken into account in the planning of agricultural development of an area. Apart from knowledge of the amount and temporal and partial pattern of the major climatic elements such as radiation, air and soil temperatures, precipitation, sunshine duration, evaporation and relative humidity which influence agriculture, knowledge of derived climatic indices from water and heat balance studies is also required. For example, through water balance studies we can obtain information on the water relations in a given area and derive measures of actual evapotranspiration, water deficit, water surplus, soil moisture deficit and other parameters useful in agricultural decision making regarding choice of crops sowing dates, and irrigation scheduling amongst others.

Knowledge of the microclimate of crops can assist in creating better environment for the growth of crops. This can be done through

i application of irrigation water in the right amount and at the right time

ii Planting of shelter belt to reduce wind speed/velocity, reduce rate of transpiration or trap snow and in temperate region by

III increasing soil temperatures in spring by dusting with soot and

IV planting in favoured slopes to intercept more radiation ~~from~~ for crop growth.

THE USEFULNESS OF AGROCLIMATOLOGICAL STUDIES IN AGRICULTURAL PLANNING AND DEVELOPMENT CAN SUMMARIZED AS FOLLOWS

1 optimal timing of agricultural operations such as land preparation, planting, weeding, harvest, storage, marketing

ii Selection of suitable site for given crops or of suitable crops for given sites.

iii Control of crop and animal pests and diseases

iv Control and management of weather phenomena which constitute hazard to agriculture.

v Assessment or determination of the desirability or otherwise of irrigation agriculture and the irrigation needs of crops,

vi Forecast of crop yield to assist the formulation of policies relating to the procurement, distribution, imports, exports and prices of foodstuffs.

vii Forecast of impact of climate change on agricultural production and

viii The design of appropriate response strategies to ensure food security.

WEATHER HAZARD IN AGRICULTURE

Climate can act as both a resource and as a constraint in agricultural production. In examining the climate crop relationships so far, emphasis has been on the role of climate as a facilitor or a resource to ensure good crop yield. ~~we~~ now we want to examine some ~~ways~~ of the ways in which weather and climate act as constraint to agricultural production.

The growth of crop is not only depend on weather conditions, crops are in fact subject to number of weather hazard in field before the grow to maturity and are harvested. The major weather phenomena that constitute hazard to agriculture are

1 Drought
2 Hail stone
3 High wind
4 Frost

4 Excessive heat

i, Drought is a serious hazard to agriculture in both temperate and tropical region of the world.
Drought can be said to occur whenever the supply of moisture from precipitation or stored in the soil is insufficient to fulfill the optimum water need of crops.
Thornthwaite (1944) has recognized four (4) types of drought namely i, permanent ii, seasonal iii, contigent iv, invisible drought.

i, permanent drought occurs in arid areas where in no season is precipitation enough to satisfy the water needs of crops. in such areas, agriculture is impossible with irrigation through the whole crop season.

ii Seasonal drought occurs in area with well-define wet and dry seasons as in most parts of the tropics. The drought occurs every year, owing to seasonal changes in atmospheric circulation pattern. Agriculture is possible during the rainy season or with the use of irrigation during the dry season.

iii Contigent drought is characteristic of subhumid and humid areas and occurs when over a period of time that expected rain fails to fall. Contigent drought is unpredictable and therefore constitutes a serious hazard to agriculture.

IV invisible drought cannot ~~be~~ easily be recognised like the other types of drought which are quite evident by the wilting of crops and lack of vegetative growth that they cause. in invisible drought, the crops do not wilt but fail to grow at their optimal rates because the daily supply of moisture from the soil or through falling precipitation does not match the daily water requirement of crops.

2 Hailstone = Hailstone cause physical damage to Crops. They are hard pellets of ice of various size and shapes that falls from cumulonimbus clouds. Hailstones occur both in the temperate region and in the tropics. Crop losses due to hailstone may be considerable and so various attempt are made by farmers to protect their crops against the hazard. In medieval Europe, people used to shoot arrows into approaching storms or ring church bell in the mistaken belief that these would prevent hail from forming and falling. Nowadays, crop losses due to hailstone are prevented seeding cloud with silver iodide (or dry ice) released from ~~artillery~~ artillery shell or aircraft. The objective of seeding is to create more in size ice particles and ensure that hailstone that form are very small in size and less damaging to crops.

2 High wind – wind transports moisture and heat in the atmosphere and consequently exert some influence on crop production. Wind also influences the rates of evapotranspiration and directly exert pressure on crops along the path.

Wind may constitute hazard to agriculture in one or more of the following:

i Wind may phisically damage crops

ii Hot wind encourage high rate of evapotranspiration which may cause desiccation in crops.

iii Hot wind increase the risk of fire which destroy crops.

IV Wind encourages soil erosion that removes the top soil and generates dust that may damage crops by abrasion by sand particles or by covering them up.

V Cold wind may cause chilling of crops.

Damage to crops by wind may be minimized or prevented by the use of windbreaks, also known as shelterbelt.

5

4 ~~Def~~ Excessive heat 2: Excessive heat is a great hazard to germination of seeds particularly in warm environments. The effect of excessive heat can be controlled by the use of mulches. Mulches help to reduce the direct impact of solar radiation

(5) Frost is a deposite of small white ice crystals formed on the ground or other surfaces when the temperature falls below freezing. Frost damage occurs when ice forms inside the plant tissue, damaging its cells and producing a negative effect on yields, as well as depreciation in the quality of the product.